

Theory of Finite-Horizon Markov Decision Processes

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Finite-Horizon Markov Decision Processes



Finite-Horizon MDP

Let $H \in \mathbb{N}$ and define $[H] := \{1, \dots, H\}$.

A finite-horizon MDP with (fixed) horizon H is a tuple

$$M = \left(\mathcal{S}, \mathcal{A}, \{P_h\}_{h \in [H]}, \{R_h\}_{h \in [H]} \right)$$

- **State-space $\mathcal{S}$** (with size S)
- **Action-space $\mathcal{A}$** (with size A)
- **Transition function P_h** : Selecting $a \in \mathcal{A}$ in $s \in \mathcal{S}$ at time step $h \in [H]$ leads to a transition to s' with probability $P_h(s'|s, a)$.
 $P_h(\cdot|s, a)$ is a probability distribution over $\mathcal{S}$, i.e.,

$$\sum_{s'} P_h(s'|s, a) = 1$$

- **Reward function R_h** : Selecting $a \in \mathcal{A}$ in $s \in \mathcal{S}$ at time step $h \in [H]$ yields a reward $r \sim R_h(s, a)$.

- The action-space may generally be state-dependent; if necessary, we use $\mathcal{A}_s$ to denote the set of actions available in state s .
- In general, $\mathcal{S}$ or $\mathcal{A}$ could be finite, countably infinite, or continuous.



Interaction with MDP

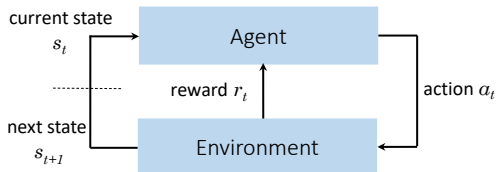
An **agent** interacts with the MDP for H rounds.

At each time step $t = 1, 2, \dots, H - 1$:

- The agent observes the current state s_t and takes an action $a_t \in \mathcal{A}$
- MDP decides a reward $r_t \sim R_t(s_t, a_t)$ and a next-state $s_{t+1} \sim P_t(\cdot | s_t, a_t)$
- The agent receives r_t (any time in step t before start of $t + 1$)

At time step $t = H$:

- The agent is in a **terminal state** s_H , receives a reward $r_H(s_H) \sim R_H(s_H)$, and takes no action.



Interaction with MDP

An **agent** interacts with the MDP for H rounds.

At each time step $t = 1, 2, \dots, H - 1$:

- The agent observes the current state s_t and takes an action $a_t \in \mathcal{A}$
- MDP decides a reward $r_t := r_t(s_t, a_t)$ and a next state $s_{t+1} \sim P_t(\cdot | s_t, a_t)$
- The agent receives r_t (any time in step t before start of $t + 1$)

At time step $t = H$:

- The agent is in a **terminal state** s_H , receives a reward $r_H(s_H)$, and takes no action.

This interaction produces a trajectory (or history)

$$h = (s_1, a_1, r_1, s_2, a_2, r_2, \dots, s_{H-1}, a_{H-1}, r_{H-1}, s_H, r_H)$$

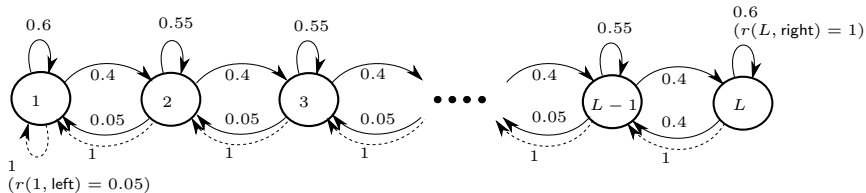
The goal is to maximize the total expected sum of rewards

$$\max_{\text{strategies}} \mathbb{E} \left[\sum_{t=1}^{H-1} r_t(s_t, a_t) + r_H(s_H) \right]$$



Example: RiverSwim

The L -state RiverSwim MDP



RiverSwim with horizon H :

- $P_1 = P_2 = \dots = P_h$
- $R_1 = R_2 = \dots = R_h$

But the model allows for having time-dependent P and R (e.g., to model changes in river dynamics or rewards).



Policy and Value Function



Policy

When interacting with an MDP, actions are taken according to some **policy**:

A policy may be:

- deterministic or randomized (stochastic)
- history-dependent, deterministic, or **Markov**

	Deterministic	Randomized
Stationary	$\pi : \mathcal{S} \rightarrow \mathcal{A}$	$\pi : \mathcal{S} \rightarrow \mathcal{A}$
Markov	$\pi : \mathcal{S} \times [H] \rightarrow \mathcal{A}$	$\pi : \mathcal{S} \times [H] \rightarrow \Delta(\mathcal{A})$
History-dependent	$\pi : \mathcal{H} \rightarrow \mathcal{A}$	$\pi : \mathcal{H} \rightarrow \Delta(\mathcal{A})$

- $\Delta(\mathcal{A})$ denotes the simplex of probability distributions over $\mathcal{A}$.
- $\mathcal{H}$ the set of all possible history sequences (trajectories).
- For a randomized policy π , $\pi(a|s)$ denotes the probability of choosing a in s .

A Markov policy only depends on the current state, but may vary over time.



Policy

- Π^{SD} : Stationary deterministic policies
- Π^{SR} : Stationary randomized policies
- Π^{MD} : Markov deterministic policies
- Π^{MR} : Markov randomized policies
- Π^{HD} : History-dependent deterministic policies
- Π^{HR} : History-dependent randomized policies

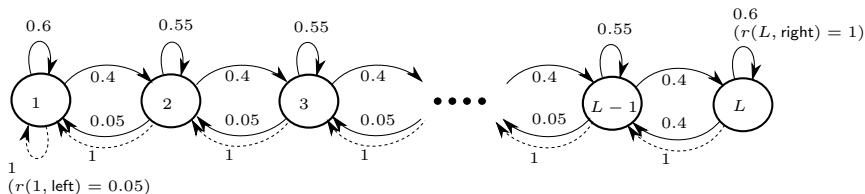
	Deterministic	Randomized
Stationary	$\pi : \mathcal{S} \rightarrow \mathcal{A}, \Pi^{\text{SD}}$	$\pi : \mathcal{S} \rightarrow \mathcal{A}, \Pi^{\text{SR}}$
Markov	$\pi : \mathcal{S} \times [H] \rightarrow \mathcal{A}, \Pi^{\text{MD}}$	$\pi : \mathcal{S} \times [H] \rightarrow \Delta(\mathcal{A}), \Pi^{\text{MR}}$
History-dependent	$\pi : \mathcal{H} \rightarrow \mathcal{A}, \Pi^{\text{HD}}$	$\pi : \mathcal{H} \rightarrow \Delta(\mathcal{A}), \Pi^{\text{HR}}$

- (i) if π is stationary $\implies \pi$ is Markov
- (ii) $\Pi^{\text{MD}} \subset \Pi^{\text{MR}} \subset \Pi^{\text{HR}}$
- (iii) $\Pi^{\text{MD}} \subset \Pi^{\text{HD}} \subset \Pi^{\text{HR}}$



Example: RiverSwim

The L -state RiverSwim MDP



RiverSwim with horizon H :

- $P_1 = P_2 = \dots = P_h$
- $R_1 = R_2 = \dots = R_h$

Policy Examples:

- π_1 : Always swim to right $\implies \pi_1 \in \Pi^{\text{SD}}$
- π_2 : Swim to right if $t < 5$, and to left if $t > 5 \implies \pi_2 \notin \Pi^{\text{SD}}, \pi_2 \in \Pi^{\text{MD}}$



Value Function

The **value function** of policy π is a mapping $V^\pi : \mathcal{S} \times [H] \rightarrow \mathbb{R}$ defined as

$$V^\pi(s, h) := \mathbb{E}^\pi \left[\sum_{t=h}^{H-1} r_t(s_t, a_t) + r_H(s_H) \middle| s_h = s \right].$$

where $\mathbb{E}^\pi$ indicates expectation over trajectories generated by π .

- $V^\pi(s, h)$ measures the sum of $H - h + 1$ future rewards (in expectation), when the agent starts in s at time h and follows π .
- For all π , s , and h :

$$|V^\pi(s, h)| \leq (H - h + 1)R_{\max},$$

where $R_{\max}$ is an upper bound on the absolute value of rewards.



Policy Evaluation



Bellman Equation for π

Theorem (Bellman Equation for $\pi \in \Pi^{MD}$)

Let $\pi \in \Pi^{MD}$. For all $s \in \mathcal{S}$,

- For all terminal states s , $V^\pi(s, H) = r_H(s)$
- For $h = H - 1, \dots, 1$, and for all non-terminal s ,

$$V^\pi(s, h) = r_h(s, \pi(s, h)) + \sum_{x \in \mathcal{S}} P_h(x|s, \pi(s, h)) V^\pi(x, h + 1)$$

- Gives a method for Policy Evaluation using $O(HS^2A)$ operations.
- Theorem holds for $\pi \in \Pi^{MR}$:

$$V^\pi(s, h) = \mathbb{E}^\pi[r_h(s, a)] + \sum_{x \in \mathcal{S}} \mathbb{E}^\pi[P_h(x|s, a) V^\pi(x, h + 1)]$$

where $\mathbb{E}^\pi[\square(a)] = \sum_{a \in \mathcal{A}} \square(a) \pi(a|s, h)$.



Bellman Equation for π

We prove the theorem for $\pi \in \Pi^{\text{MD}}$. (See Lecture Notes for $\pi \in \Pi^{\text{MR}}$.)

Proof. Let $\pi \in \Pi^{\text{MD}}$, and define $\pi_h(s) := \pi(s, h)$. We have, for $s \in \mathcal{S}$ and $h \in [H - 1]$,

$$\begin{aligned}
 V^\pi(s, h) &= \mathbb{E}^\pi \left[\sum_{t=1}^{H-1} r_h(s_t, \pi_t(s_t)) + r_H(s_H) \middle| s_h = s \right] \\
 &= r_h(s, \pi_h(s)) + \mathbb{E}^\pi \left[\sum_{t=h+1}^{H-1} r_h(s_t, \pi_t(s_t)) + r_H(s_H) \middle| s_h = s \right] \\
 &= r_h(s, \pi_h(s)) \\
 &\quad + \sum_{x \in \mathcal{S}} \mathbb{P}(s_{h+1} = x | s_h = s, a_h = \pi_h(s_h)) \underbrace{\mathbb{E}^\pi \left[\sum_{t=h+1}^{H-1} r_h(s_t, \pi_t(s_t)) \middle| s_{h+1} = x \right]}_{= V^\pi(x, h+1)} \\
 &= r_h(s, \pi_h(s)) + \sum_{x \in \mathcal{S}} P(x | s, \pi_h(s)) V^\pi(x, h+1).
 \end{aligned}$$



Optimization in Finite-Horizon MDPs: Optimal Policy and Value



Optimal Value and Policy

Solving a finite-horizon MDP M amounts to solving the following optimization problem:

$$V^*(s, 1) = \sup_{\pi \in \Pi^{\text{HR}}} V^\pi(s, 1),$$

for all $s \in \mathcal{S}$. Any policy achieving $V^*(s, 1)$ for all s is called **optimal** and denoted by π^* .

Theorem

There exists an optimal Markov deterministic policy.



Optimal Bellman Equation

Theorem

The optimal value function $V^*(s, h)$ is the solution to the *optimal Bellman equations*:

$$V^*(s, H) = r_H(s)$$

$$V^*(s, h) = \max_{a \in \mathcal{A}} \left(r_h(s, a) + \sum_{x \in \mathcal{S}} P_h(x|s, a) V^*(x, h+1) \right),$$

$$\forall h = H-1, \dots, 1$$

Optimal policy π^* : For all $s \in \mathcal{S}$ and $h = H-1, \dots, 1$,

$$\pi^*(s, h) \in \operatorname{argmax}_{a \in \mathcal{A}} \left(r_h(s, a) + \sum_x P_h(x|s, a) V^*(x, h+1) \right).$$



Dynamic Programming

- **initialization:** For all terminal states s : $V^*(s, H) = r_H(s)$
- **for** $t = H - 1, \dots, 1$
For all s

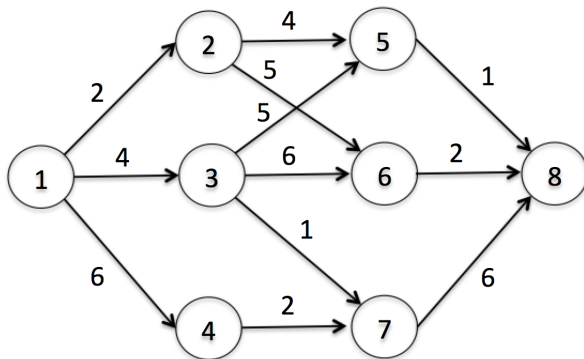
$$V^*(s, h) = \max_{a \in \mathcal{A}} \left(r_h(s, a) + \sum_{x \in \mathcal{S}} P_h(x|s, a) V^*(x, h + 1) \right)$$

$$\pi^*(s, h) \in \operatorname{argmax}_{a \in \mathcal{A}} \left(r_h(s, a) + \sum_{x \in \mathcal{S}} P_h(x|s, a) V^*(x, h + 1) \right)$$

output: π^*



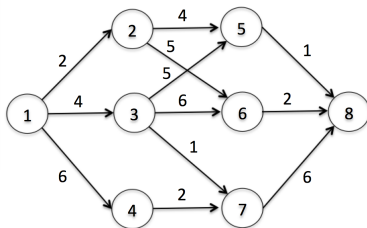
Example: Routing



Find the maximum-weight route between source (node 1) and destination (node 8).



Example: Routing



Modeling as finite-horizon MDP:

- States: Nodes in the graph $\mathcal{S} = \{1, 2, \dots, 8\}$
- Actions: Outgoing edges at each state, e.g., $\mathcal{A}_2 = \{\text{go to 4, go to 5}\}$
- Deterministic transitions
- Rewards: Edge weights
- Time horizon N : or any number greater than the maximum path length (here, $N \geq 4$)

The full solution is provided in the lecture note.

